

Preconditioning without a Preconditioner

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Paper

Preconditioning without a preconditioner using randomized block Krylov subspace methods

Tyler Chen, Caroline Huber, Ethan Lin, Hajar Zaid

<https://arxiv.org/abs/2501.18717> (to appear in ETNA)

Context

I am broadly interested in understanding the following:

Why don't we use block-KSMs for matrix functions?

Introduction

We are interested in solving the PSD linear system

$$\mathbf{A}_\mu \mathbf{x} = \mathbf{b}, \quad \mathbf{A}_\mu := \mathbf{A} + \mu \mathbf{I}.$$

This is relevant for:

- Solving PSD linear systems
- Ridge regression (Tikhonov regularization)
- Sampling Gaussian vectors

In the latter two examples, we might need the solution for many values μ efficiently.

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In the latter two examples, we might need the solution for many values μ efficiently.

Preconditioned Conjugate Gradient

CG produces an optimal solution over the Krylov subspace

$$\mathcal{K}_t(\mathbf{A}, \mathbf{b}) := \text{span}\{\mathbf{b}, \mathbf{A}\mathbf{b}, \dots, \mathbf{A}^{t-1}\mathbf{b}\}.$$

If $\mathbf{A}_\mu$ is poorly conditioned, then CG might converge slowly. So it is common to work over a “preconditioned Krylov subspace”:

$$\mathcal{K}_t(\mathbf{A}_\mu, \mathbf{b}; \mathbf{P}_\mu) := \mathcal{K}_t(\mathbf{P}_\mu^{-1}\mathbf{A}_\mu, \mathbf{P}_\mu^{-1}\mathbf{b}).$$

PCG converges in

$$\sqrt{\kappa(\mathbf{P}_\mu^{-1/2}\mathbf{A}_\mu\mathbf{P}_\mu^{-1/2})} \log\left(\frac{1}{\varepsilon}\right) \text{ iterations.}$$

Need to choose preconditioner that is easy to construct and apply but also reduces condition number.

Deflation Preconditioner

Often, $\mathbf{A}_\mu$ is ill-conditioned because of r large eigenvalues: $\lambda_r \gg \lambda_{r+1} \approx \lambda_d$.

One can use the “deflation preconditioner”

$$\mathbf{P}_\mu := \frac{1}{\theta + \mu} \mathbf{U}(\mathbf{D} + \mu \mathbf{I}) \mathbf{U}^T + (\mathbf{I} - \mathbf{U} \mathbf{U}^T),$$

where $\mathbf{U} \mathbf{D} \mathbf{U}^T$ is the eigendecomposition of the best rank- r approx to $\mathbf{A}$.

Top eigenvalues mapped to $\theta + \mu$, bottom eigenvalues untouched.

So if $\theta \in [\lambda_d + \mu, \lambda_{r+1} + \mu]$, we get convergence in

$$\sqrt{\kappa_r(\mathbf{A}_\mu)} \log \left(\frac{1}{\varepsilon} \right) \text{ iterations.}$$

Fast convergence!

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Fast convergence!

Nyström low-rank approximation

Computing top eigenvectors is hard! However, RandNLA gives us very good approximate methods.

When $\mathbf{A}$ is positive semi-definite, a good choice is the Nyström approximation

$$\mathbf{A}\langle\mathbf{K}\rangle := (\mathbf{AK})(\mathbf{K}^T\mathbf{AK})^{-1}(\mathbf{K}^T\mathbf{A}).$$

We can build a preconditioner by taking $\mathbf{UDU}^T$ as the eigendecomposition of $\mathbf{A}\langle\mathbf{K}\rangle$.

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Nyström PCG²

Proposition.¹ Let $\mathbf{P}_\mu$ be the Nyström preconditioner corresponding to the Nyström approximation $\mathbf{A}\langle\mathbf{K}\rangle$ for any $\mathbf{K}$ and shift parameter $\theta \geq 0$. Then

$$\kappa(\mathbf{P}_\mu^{-1/2}\mathbf{A}_\mu\mathbf{P}_\mu^{-1/2}) \leq (\theta + \mu + \|\mathbf{A} - \mathbf{A}\langle\mathbf{K}\rangle\|) \left(\frac{1}{\theta + \mu} + \frac{1}{\lambda_d + \mu} \right).$$

If $\|\mathbf{A} - \mathbf{A}\langle\mathbf{K}\rangle\| \approx \lambda_{r+1}$, then Nyström PCG behaves similar to deflation!

Caveat: The space $\mathcal{K}_t(\mathbf{A}_\mu, \mathbf{b}; \mathbf{P}_\mu)$ depends on μ ! So we need to re-run PCG for each value of μ . (in contrast, $\mathcal{K}_t(\mathbf{A}_\mu, \mathbf{b})$ does not depend on μ)

¹Frangella, Tropp, and Udell 2023.

²Martinsson and Tropp 2020; Frangella, Tropp, and Udell 2023; Carson and Daužickaitė 2024; Díaz et al. 2023; Zhao et al. 2024; Hong et al. 2024; Dereziński, Musco, and Yang 2025.

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Nyström PCG bounds

Theorem.³ Suppose $\mathbf{K}$ is a Gaussian sketching matrix. If the sketching dimension is on the order of

$$d_{\text{eff}}(\mu) := \text{tr}(\mathbf{A}\mathbf{A}_{\mu}^{-1}) = \sum_{i=1}^d \frac{\lambda_i}{\lambda_i + \mu},$$

then Nyström PCG converges in

$$\log \left(\frac{1}{\varepsilon} \right) \text{ iterations.}$$

Good: No dependence on the condition number!

To improve: How do we know the effective dimension? What if $\mu = 0$? What if there are multiple μ ?

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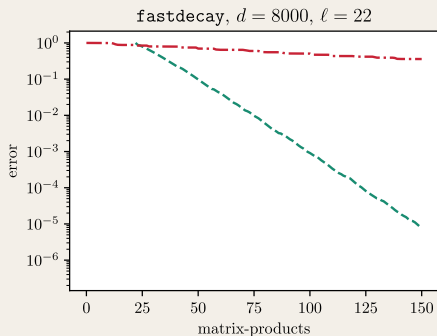
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Example

Test problem with a few outlying eigenvalues.



Legend: CG (—·—) and Nyström PCG with $s = 1$ (- - -).

Block Krylov Nyström

As a simple generalization, we consider block Krylov Nyström

$$\mathbf{A}\langle \mathbf{K}_s \rangle := (\mathbf{A}\mathbf{K}_s)(\mathbf{K}_s^T \mathbf{A}\mathbf{K}_s)^{-1}(\mathbf{K}_s^T \mathbf{A})., \quad \mathbf{K}_s := [\mathbf{G} \ \mathbf{A}\mathbf{G} \ \dots \ \mathbf{A}^{s-1}\mathbf{G}].$$

Theorem.⁴ Suppose $\mathbf{G} \in \mathbb{R}^{d \times (r+p)}$ is a random Gaussian matrix. Then, if $p \geq 2$,

$$\log \left(\frac{\mathbb{E} \|\mathbf{A} - \mathbf{A}\langle \mathbf{K}_s \rangle\|^2}{\lambda_{r+1}^2} \right) \leq \frac{1}{8(s - \frac{3}{2})^2} \log \left(4 + \frac{4r}{p-1} \sum_{i>r} \frac{\lambda_i^2}{\lambda_{r+1}^2} \right)^2.$$

This let's us prove a bound that says that if $s \approx \log(d)$, we get convergence in $\sqrt{\kappa_r(\mathbf{A}_\mu)} \log(1/\varepsilon)$ iterations.

To improve: Still don't know how to set block size $r + p$ or handle multiple μ .

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Block Krylov Methods

Given a matrix $\mathbf{B} \in \mathbb{R}^{d \times m}$ (typically $m \ll d$) with columns $\mathbf{b}^{(1)}, \dots, \mathbf{b}^{(m)}$, the block Krylov subspace is defined as

$$\mathcal{K}_t(\mathbf{A}, \mathbf{B}) := \mathcal{K}_t(\mathbf{A}, \mathbf{b}^{(1)}) + \dots + \mathcal{K}_t(\mathbf{A}, \mathbf{b}^{(m)}).$$

In at least some settings,⁵ matrix-loads/iterations/matrix block-vector products are the dominant cost, in which case building a block Krylov subspace is nearly the same cost as building a single Krylov subspace.

⁵If it not realistic in your setting, please wait until the end of the talk to complain.

Block CG

Block CG is a generalization of CG that is optimal over the block Krylov subspace.

- If we care about $\mathbf{Ax} = \mathbf{b}^{(1)}$, block CG is no worse than CG (in terms of matrix-loads) because $\mathcal{K}_t(\mathbf{A}, \mathbf{b}^{(1)}) \subseteq \mathcal{K}_t(\mathbf{A}, \mathbf{B})$.

High-level question: When is block-CG better than CG?

Silly observation⁶

Theorem. Suppose $\mathbf{P}_\mu = (\mathbf{I} + \mathbf{X})^{-1}$, where $\text{range}(\mathbf{X}) \subseteq \mathcal{K}_{s+1}(\mathbf{A}_\mu, \mathbf{G})$. Then,

$$\mathcal{K}_t(\mathbf{A}_\mu, \mathbf{b}; \mathbf{P}_\mu) \subseteq \mathcal{K}_t(\mathbf{A}, \mathbf{b}) + \mathcal{K}_{t+s}(\mathbf{A}, \mathbf{G}).$$

Proof. By definition, $\mathcal{K}_t(\mathbf{A}_\mu, \mathbf{b}; \mathbf{P}_\mu)$ consists of linear combinations of the vectors

$$(\mathbf{P}_\mu^{-1} \mathbf{A}_\mu)^k \mathbf{P}_\mu^{-1} \mathbf{b} = ((\mathbf{I} + \mathbf{X}) \mathbf{A}_\mu)^k (\mathbf{I} + \mathbf{X}) \mathbf{b}, \quad k = 0, 1, \dots, t-1,$$

and each $((\mathbf{I} + \mathbf{X}) \mathbf{A}_\mu)^k (\mathbf{I} + \mathbf{X}) \mathbf{b}$ can be expressed as linear combination of vectors which live in the specified space.

⁶Chen, Huber, et al. 2026.

Silly main theorem⁷

Theorem. Fix any matrix $\mathbf{G} \in \mathbb{R}^{d \times m}$ and let $\mathbf{P}_\mu = (\mathbf{I} + \mathbf{X})^{-1}$ be any preconditioner where $\text{range}(\mathbf{X}) \subseteq \mathcal{K}_{s+1}(\mathbf{A}, \mathbf{G})$. Define the augmented starting block $\mathbf{B} = [\mathbf{b} \ \mathbf{G}]$. Then, for any $t \geq s$, the t -th block-CG iterate is related to the $(t - s)$ -th preconditioned-CG iterate corresponding to the preconditioner $\mathbf{P}_\mu$ in that

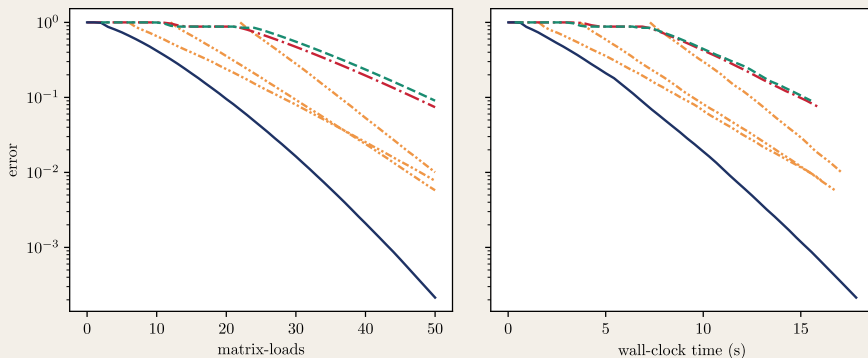
$$\|\mathbf{A}_\mu^{-1} \mathbf{b} - \text{bcg}_t^{(1)}(\mu)\|_{\mathbf{A}_\mu} \leq \|\mathbf{A}_\mu^{-1} \mathbf{b} - \text{pcg}_{t-s}(\mu)\|_{\mathbf{A}_\mu}.$$

In short: augmented block-CG automatically performs no worse than Nyström PCG (with the best choice of s and θ) after the **same number of matrix-loads**.

- Our new bounds for Nyström PCG immediately give new bounds for augmented block CG (where $\mathbf{G}$ is Gaussian).

⁷Chen, Huber, et al. 2026.

Example



Legend: ours (—), standard CG (-.-), Nyström PCG from Frangella, Tropp, and Udell 2023 (---), and generalizations Nyström PCG using larger s (.....).

What are the costs?

For Nyström PCG:

- Nyström approximation with starting block size ℓ requires s matrix block-vector products with $\mathbf{A}$ (each requires ℓ matvecs)
- Each iteration of PCG requires 1 matvec with $\mathbf{A}$
- Unclear how to choose s, ℓ .
- We require a new preconditioner for every value of μ .

For block CG:

- Each iteration of BCG requires 1 block-vector products with $\mathbf{A}$
- Unclear how to choose ℓ .
- Very cheap to solve for multiple values of μ .

What about the block-size?

So far: set $\ell = O(r)$ and run for at least $s = \log(d)$ iterations.

But what if we don't know r ahead of time!?

- Since $\mathbf{A}\langle\mathbf{K}_s\rangle$ has rank as large as ℓs , we might hope that things are okay as long as $\ell s = O(r)$.

Theorem.⁸ Let $\mathbf{A} \in \mathbb{R}^{d \times \ell}$ be a Gaussian matrix. Then, for some

$$s = O\left(\frac{r}{\ell\sqrt{\varepsilon}} \log(\Delta) + \log\left(\frac{d}{\delta\varepsilon}\right)\right), \quad \Delta := \min_{i=1,\dots,\ell\lceil r/\ell\rceil-1} \frac{\lambda_i - \lambda_{i+1}}{\lambda_1},$$

with probability at least $1 - \delta$, there is a matrix $\mathbf{Q} \in \mathbb{R}^{d \times r}$ with orthonormal columns and $\text{range}(\mathbf{Q}) \subseteq \mathcal{K}_s(\mathbf{A}, \cdot)$ such that

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\| < (1 + \varepsilon)\lambda_{r+1}.$$

Then easily extend this bound to Nystrom.

⁸Chen, Epperly, et al. 2025.

Sampling Gaussians

Some applications ask to sample from $\mathcal{N}(\mathbf{m}, \mathbf{A})$. This can be done by

$$\mathbf{m} + \mathbf{A}^{1/2}\mathbf{b} \sim \mathcal{N}(\mathbf{m}, \mathbf{A}), \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

There is a nice identity:

$$\mathbf{A}^{1/2}\mathbf{b} = \frac{2}{\pi} \int_0^\infty \mathbf{A}(\mathbf{A} + z^2\mathbf{I})^{-1}\mathbf{b} \, dz.$$

At each point z , we need to solve a shifted linear system $(\mathbf{A} + z^2\mathbf{I})\mathbf{x} = \mathbf{b}$.

Using augmented block-CG to analyze Block Lanczos-FA

Suppose we want to sample m Gaussians. We need to compute $\mathbf{A}^{1/2}\mathbf{b}^{(1)}, \dots, \mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{b}^{(m)}$, where $\mathbf{b}^{(1)}, \dots, \mathbf{b}^{(m)}$ be standard Gaussian vectors.

On the last slide, we saw how to reduce this to solving

$$(\mathbf{A} + z^2\mathbf{I})\mathbf{x} = \mathbf{b}^{(1)}.$$

Note that $[\mathbf{b}^{(2)}, \dots, \mathbf{b}^{(m)}]$ is a Gaussian matrix independent of $\mathbf{b}^{(1)}$! So we can use augmented BCG with these vectors as the Gaussian sketch.

Same argument for $\mathbf{b}^{(2)}, \dots, \mathbf{b}^{(m)}$.

But the block Krylov subspace is the same for all of them, so we only need to build it once!

Convergence guarantees

High-level question: when do block KSMs for matrix functions outperform running a single-vector algorithm on each vector?

Lanczos-FA:

$m\sqrt{\kappa} \log(1/\varepsilon)$ matrix-vector products.

Block Lanczos-FA:

$m \left(\log(d) + \sqrt{\kappa_{r+1}} \log(\log(\kappa)/\varepsilon) \right)$ matrix-vector products,

where $r = O(m / \log(m))$.

Use in Gaussian Process regression

In certain applications, we need to compute $\mathbf{A}^{-1}\mathbf{b}$ and $\text{tr}(\log(\mathbf{A}))$.

If we use Girard–Hutchinson estimator for the trace and Lanczos for the matrix function, then we end up a block Krylov subspace on random vectors. And people have been appending on $\mathbf{b}$ and then using this augmented block Krylov subspace.⁹.

⁹Gardner et al. 2018; Xu et al. 2025.

Back to a single system: let's complicate the story

Earlier, I asked you not to complain about my access model (counting block-matvecs).¹⁰

The problem is, the story is actually kind of subtle, and there is no clear winner...

¹⁰The referees definitely did complain about this :/

Exact arithmetic CG

In exact arithmetic, vanilla CG already satisfies bounds in terms of $\kappa_r(\mathbf{A}_\mu)$.

Theorem. For any $r \geq 0$ the CG iterate satisfies

$$\frac{\|\mathbf{A}_\mu^{-1}\mathbf{b} - \text{cg}_t(\mu)\|_{\mathbf{A}_\mu}}{\|\mathbf{A}_\mu^{-1}\mathbf{b}\|_{\mathbf{A}_\mu}} \leq 2 \exp\left(-\frac{2(t-r)}{\sqrt{\kappa_{r+1}(\mu)}}\right).$$

Proof. Recall the CG error can be bounded in terms of

$$\min_{\substack{\deg(p)=t \\ p(0)=1}} \left(\max_{x \in \text{spec}(\mathbf{A})} |p(x)| \right).$$

To get the usual bound, we relax to $[\lambda_d, \lambda_1]$ and use a shifted and scaled Chebyshev polynomial.

Instead, relax to $[\lambda_d, \lambda_{r+1}] \cup \{\lambda_r, \dots, \lambda_1\}$, and place explicit zeros at $\lambda_r, \dots, \lambda_1$.

Finite precision arithmetic

We can use the previous bound to prove a bound in terms of the effective dimension

$$d_{\text{eff}}(\mu) := \text{tr}(\mathbf{A}\mathbf{A}_{\mu}^{-1}) = \sum_{i=1}^d \frac{\lambda_i}{\lambda_i + \mu}.$$

Theorem.¹¹ The CG iterate satisfies

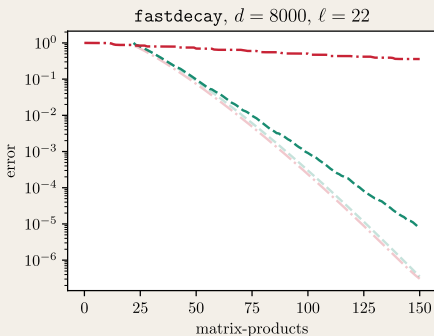
$$\frac{\|\mathbf{A}_{\mu}^{-1}\mathbf{b} - \text{cg}_t\|_{\mathbf{A}_{\mu}}}{\|\mathbf{A}_{\mu}^{-1}\mathbf{b} - \text{cg}_0\|_{\mathbf{A}_{\mu}}} \leq 2 \exp\left(-\sqrt{2}(t - 2d_{\text{eff}}(\mu))\right).$$

Takeaway: The value of Nyström PCG over CG (in terms of matvecs) is predicated on being in finite precision arithmetic.

¹¹Has anyone seen this bound before? It is immediate from the fact that $r > 2d_{\text{eff}}(\mu) \implies \lambda_{r+1} \leq \mu$.

Example

Test problem with a few outlying eigenvalues (light=full reorth).



Legend: CG (—·—) and Nyström PCG with $s = 1$ (- - -)

What if CG gets a little orthogonalization?

Full reorthogonalization might be too expensive.

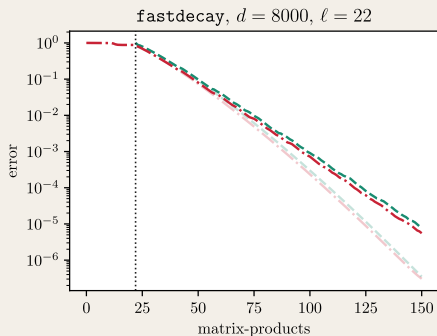
However:

- to build the Nyström preconditioner, we needed to orthogonalize ℓ vectors
- to apply the Nyström preconditioner, we need to compute ℓ inner products

If we run CG with orthogonalization for ℓ steps, and then only orthogonalize against these vectors, the computational profile is very similar to Nyström PCG.

Example

Let CG orthogonalize for the first ℓ steps.



Legend: CG (—·—) and Nyström PCG with $s = 1$ (- - -).

Discussion

- CG doesn't need to set ℓ in advance and get sets similar performance. So does this mean Nyström PCG is pointless?
 - Not necessarily. Because the ℓ matvecs used in the sketch can be done in parallel (whereas CG does them sequentially).
- But if this is cheaper than doing ℓ separate matvecs, then you need to start thinking about augmented block-CG.
 - And the same ideas don't require you to build the entire block Krylov subspace, you can try to do some kind of deflation.

My take: I don't really think augmented block-CG in the form described here it is a good idea in most cases. But the overall idea of just building Krylov subspaces instead of preconditioners is promising.

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